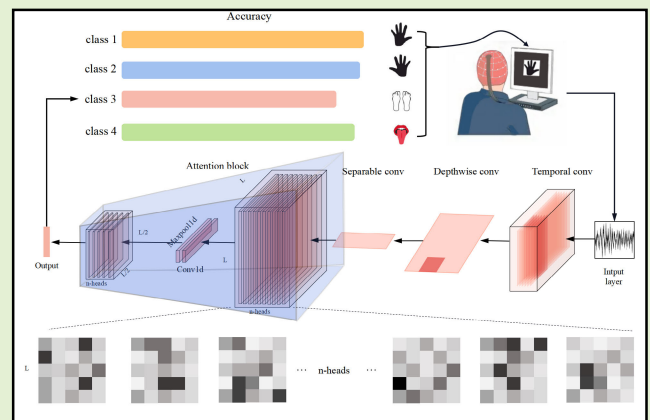


EEGNet-MSD: A Sparse Convolutional Neural Network for Efficient EEG-Based Intent Decoding

Rongrong Fu¹, Zeyi Wang¹, Shiwei Wang¹, Xuechen Xu, Junxiang Chen², and Guilin Wen

Abstract—Electroencephalography (EEG) is a noninvasive technique that can be used in brain machine interface (BMI) systems to measure and record brain electrical activity. Deep learning (DL) techniques have proved superior to conventional methods in EEG-based intent decoding. However, some DL models have overly complex structures while ensuring the accuracy of EEG recognition, resulting in reduced training and recognition speed. In this study, we proposed a compact multihead self-attention DL decoder that combined the convolutional neural network (CNN)-based EEGNet decoder with the ProbSparse multihead self-attention mechanism. Compared with traditional self-attention methods, this decoder ensures alignment dependent on both time complexity and memory usage of $O(L \log L)$ and it has been demonstrated to enhance the accuracy of EEG-based intent recognition. The test results on dataset 2a from BCI Competition IV showed that the EEGNet multihead self-attention decoding (EEGNet-MSD) decoder performed approximately 8% better than the competition-winning decoder filter bank common spatial pattern (FBCSP) and namely batch and pairwise (NBPW), and achieved better results than the latest long short-term memory (LSTM) neural decoding method. In addition, a binary classification test was performed on the Physiobank EEG motor imagery (MI) dataset, and the results showed that the accuracy of EEGNet-MSD was approximately 4% higher than EEGNet, validating the stability of the EEGNet-MSD decoder. This study provides a new solution for enhancing the performance of EEG-based intent decoding in both accuracy and decoding speed.

Index Terms—Convolutional neural network (CNN), electroencephalography (EEG), intent decoding, motor imagery (MI), ProbSparse self-attention.



I. INTRODUCTION

IN RECENT years, the application of brain machine interface (BMI) has gradually realized the interaction between

humans and external devices. This technology controls external devices by monitoring brain activity. The best-known paradigm of BMI is motor imagery (MI), which refers to the act of imagining a movement without executing it. Subjects imagine limb movements during an MI experiment, which generates electroencephalography (EEG) signals that can be collected through noninvasive methods. Decoding the collected EEG signals is complex and challenging due to their nonlinear and nonstationary characteristics and low spatial resolution [1]. The effectiveness of EEG signals decoding methods directly affects the performance of the BMI system. Thus, if a system can accurately recognize the brain signal patterns associated with MI, then patients who lack motor function can take advantage of neuroprosthetic devices such as prostheses, orthoses, exoskeletons, and others that can be controlled through the mind [2].

In BMI systems, statistical decoding methods include empirical wavelet transform [3] and Kalman filtering [4], and so on. Although these traditional methods can effectively decode EEG signals by obtaining features, the computational process is relatively complex. As deep learning (DL) can reduce the workload of extracting features for decoding, it can serve as a

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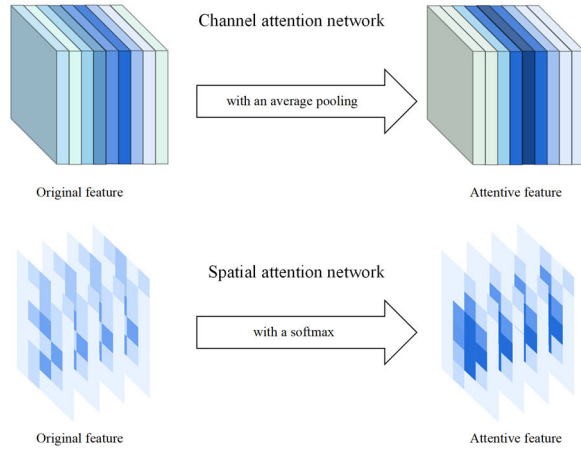


Fig. 1. Impact of channel and spatial attention networks on feature maps.

decoder in BCI systems, providing more concise and efficient computation. Several studies have demonstrated the feasibility of DL in EEG intent recognition [5], [6], [7], [8]. Many decoders based on convolutional neural networks (CNNs) have successfully decoded the intent of EEG signals [9], [10]. Lawhern et al. [11] proposed a lightweight CNN named EEGNet, which encapsulates the concept of EEG feature extraction. Using a small-dimensional neural network reduces the number of parameters used in the network, resulting in a good performance in identifying the intent of EEG signals. Several studies have reported better performance by combining EEGNet with existing models [12], [13], [14], [15], [16]. As with other neural network techniques, the attention mechanism aims to replicate how the human brain and vision process information. This approach assigns higher weights to relevant parts and lower weights to irrelevant details, enabling the network to focus more accurately and efficiently on the essential aspects rather than processing the entire feature space [17], [18]. The contrasts in feature maps under a simple channel and spatial attention network are illustrated in Fig. 1. The channel attention network aims to boost the channel in the feature map that conveys more important information and silences the other channels; the spatial attention network highlights regions of interest in the feature space and covers up the background regions.

Instead of performing a single attention function, Duan et al. [19] found it beneficial to capture different contexts with multiple individual attention functions. The multihead self-attention mechanism can compute feature weights, allowing the model to assign corresponding weights to different channels. An example of a three heads attention model is shown in Fig. 2. For the fifth query, orange, green, and blue heads focus on the different positions of the third, second, and seventh keys, respectively. In addition, each column corresponds to the channel.

In a recent study, Zhou et al. [20] proposed a ProbSparse self-attention mechanism. The advantage of this mechanism is to achieve $O(L \log L)$ time complexity and $O(L \log L)$ memory usage that depends on alignment. The ProbSparse mechanism has significantly improved compared to traditional self-attention mechanisms that typically have a quadratic

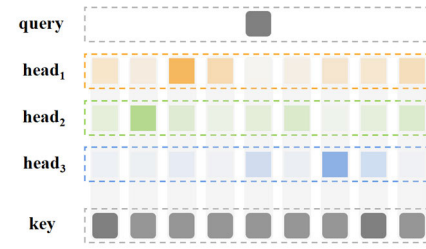


Fig. 2. Example of multihead attention (darker colors indicate a higher probability of attention being directed toward the corresponding subspace).

time complexity and require much memory. The distillation operation in the ProbSparse self-attention mechanism reduces the overall space complexity to $O((2 - \epsilon)L \log L)$, enabling faster extraction of important features, particularly suitable for handling long sequence inputs, such as EEG signals. Compared to other time series prediction methods, such as decoding cortical activity using long short-term memory (LSTM) recurrent networks [21], ProbSparse exhibits superior performance. In this study, we propose a neural network decoder for EEG intent decoding that combines the two main modules of EEGNet with the ProbSparse self-attention mechanism. This decoder is capable of extracting features, calculating attention weights, refining the features through distillation operations, and producing classification results at the feature classification layer. The EEGNet multihead self-attention decoding (EEGNet-MSD) model was compared with existing decoding methods such as the LSTM neural decoder on the BCI Competition IV dataset 2a, using accuracy and kappa value as evaluation metrics for the decoder. Additionally, experiments were conducted on the Physiobank EEG motor movement/MI dataset to validate the stability of the EEGNet-MSD model in decoding training across different datasets.

II. MATERIALS AND METHODS

A. Description of Datasets

1) *BCI Competition IV Dataset 2a*: The BCI Competition IV dataset 2a has two separate MI tasks on different dates and was provided by the Graz University of Technology in Austria. In the tasks, nine participants were instructed to imagine movements of their left hand, right hand, both feet, and tongue to gather EEG information. Each MI task has 72 experiments lasting for 7.5 s, generating 288 EEG records. Twenty-two EEG channels and three electrooculography (EOG) channels were used to record EEG signals. The signal was processed using a bandpass filter between 0.5 and 100 Hz and a notch filter at 50 Hz to suppress the noise. Data from the fourth participant was excluded from the analysis due to issues with the EOG signal recordings.

2) *Physiobank EEG Motor Movement/Imagery Dataset*: Evaluate the stability of the EEGNet-MSD decoder on the Physiobank EEG motor movement/MI dataset, which includes EEG signals from 109 individuals performing motor imagery tasks [22]. In this study, only experimental data with participants imagining their left and right wrists open or closed were selected as the dataset. The participants completed 45 trials

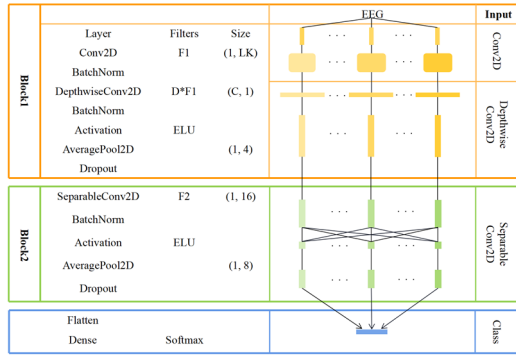


Fig. 3. Depicts the architecture of the convolution-based standard EEGNet.

across three stages, each lasting 4 s and only imagining one movement. The EEG signals were recorded from 64 channels using the international 10-10 system and the BCI2000 toolkit, and the sampling rate was 160 Hz.

B. EEGNet

EEGNet is a compact CNN for classifying EEG signals proposed by Lawhern et al. [11]. The network integrates spatial and temporal filtering notions within a CNN framework. In addition, EEGNet employs depthwise separable convolution to identify EEG features, reducing the use of parameters number in the neural network. The basic architecture of EEGNet is illustrated in Fig. 3, which can be divided into two modules. Specifically, the first module comprises two convolutional layers that capture the feature maps of input EEG signals with various bandpass frequencies [23]. In which, F1 corresponds to the number of time filters, LK denotes the size of the initial layer convolution kernel that determines the temporal convolution length, D represents the number of spatial filters, and C represents the number of channels in the EEG signal. Normalization of the data and reduction of trainable parameters is performed by the BatchNorm technique after each convolutional layer [24]. Separable convolution operations, which include depthwise and pointwise convolution layers, are utilized in the second module as an alternative to the conventional square convolution method. In Block2, F2 refers to the number of point filters. Both modules utilize an AveragePool2D layer to lower the signal sampling rate and employ Dropout to alleviate overfitting [25]. The Softmax function is used in the classification module to transform the output of Block2, ultimately producing the probabilities of each category of intent. This network has demonstrated proficient ability in categorizing EEG signals and has been confirmed across various datasets [2], [26].

C. ProbSparse Self-Attention Mechanism

ProbSparse self-attention achieves attention weight calculation by allowing each key to only attend dominant queries, reducing the time and space complexity for weight calculation. The weights calculation equation is given as follows:

$$\mathcal{A}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\tilde{\mathbf{Q}}\mathbf{K}^\top}{\sqrt{d}}\right)\mathbf{V} \quad (1)$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are the inquiry, key, and value matrices, respectively, and d represents the key and inquiry vectors'

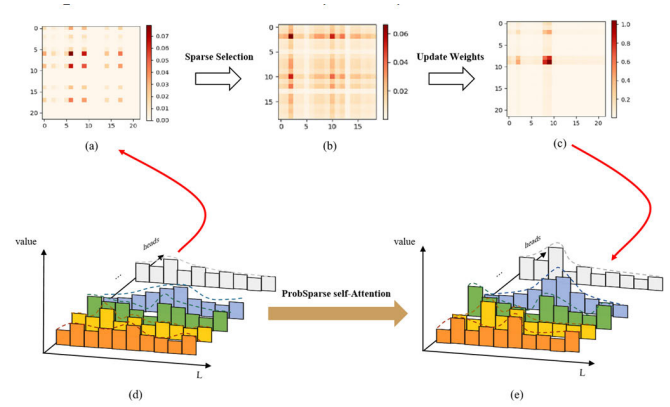


Fig. 4. Visualization of the ProbSparse self-attention mechanism. (a) Weights of features in the last attention head are computed using dot product operations. (b) Selecting a certain number for weight updates. (c) Final updated feature weights. (d) Feature values before processing using the ProbSparse self-attention mechanism. (e) Feature values after processing using the ProbSparse self-attention mechanism.

dimensional factor. is a sparse matrix, which only includes queries for the top u maximum values calculated under sparse measurement $\tilde{M}(q_i, K)$. The average value of unsampled features is calculated and replaced to reduce computational costs. Setting the sampling factor to c , the calculation for the number of samples obtained is as follows:

$$u = c \cdot \ln L_Q \quad (2)$$

where u is the number of features attended by the attention mechanism. The softmax function normalizes attention weights, and in the case of fixed sampling factor, the ProbSparse self-attention mechanism only needs to calculate $O(\ln L_Q)$ dot products per query key lookup, with a memory footprint of $O(L_k \ln L_Q)$. To achieve sparsity, an empirical approximation method called maximum mean statistics is employed

$$\tilde{M}(q_i, K) = \max_j \left\{ \frac{q_i k_j^\top}{\sqrt{d}} \right\} - \frac{1}{L_K} \sum_{j=1}^{L_K} \frac{q_i k_j^\top}{\sqrt{d}}. \quad (3)$$

The method implemented sets $L_Q = L_k = L$, where L_Q , L_k , and L represents the length of the inquiry matrix, key matrix, and value matrix, respectively. Let q_i , k_j stand for the i th row in $\mathbf{Q}$, $\mathbf{K}$, respectively. Calculating $\tilde{M}(q_i, K)$ requires randomly sampling u dot products, with the remaining pairs filled with zeros. Then, selecting sparse Top- u from them as $\tilde{\mathbf{Q}}$. As a result, the overall time and space complexity of ProbSparse self-attention is $O(L \ln L)$.

The ProbSparse self-attention mechanism is shown in Fig. 4. As shown in Fig. 4(d) and (e), they represent the changes in feature values before and after processing using the ProbSparse self-attention mechanism. The weights of features in each attention head are computed using dot product operations, shown as Fig. 4(a) for the last attention head. Next, based on a sparsity factor c , depicted as selecting a certain number based on the number of features, such as Fig. 4(b) selecting $u = 20$ out of 23 features for weight updates. Finally, the updated feature weights are obtained as shown in Fig. 4(c). The detailed implementation steps of ProbSparse self-attention are summarized in Algorithm 1.

Algorithm 1 ProbSparse Self-Attention**Require:** $Q \in \mathbb{R}^{L_Q \times d}$, $K \in \mathbb{R}^{L_K \times d}$, $V \in \mathbb{R}^{L_V \times d}$

- 1: print set hyperparameter c , $u = c \ln L_Q$ and $U = L_Q \ln L_K$
- 2: randomly select U dot-product pairs from K as $\tilde{K}$
- 3: set the sample score $\tilde{S} = Q \tilde{K}^T$
- 4: compute the measurement $M = \max(\tilde{S}) - \text{mean}(\tilde{S})$ by row
- 5: set Top- u queries under M as $\tilde{Q}$
- 6: set $S_1 = \text{softmax}(\tilde{Q} K^T / \sqrt{d}) \cdot V$
- 7: set $S_0 = \text{mean}(V)$
- 8: set $S = \{S_1, S_0\}$ by their original rows accordingly

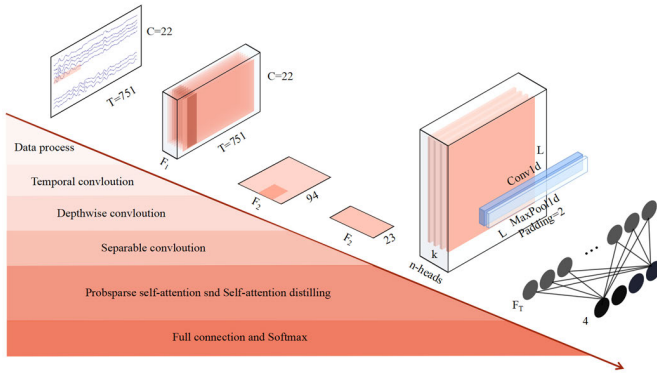


Fig. 5. Architecture of EEGNet-MSD decoder.

The self-attention distillation mechanism reduces the time dimension and advances the convolution operation through lower-level focused self-attention features. The calculation method is given as follows:

$$X_{j+1}^t = \text{MaxPool}\left(\text{ELU}\left(\text{Conv1d}\left([X_j^t]_{AB}\right)\right)\right) \quad (4)$$

where, $[]_{AB}$ represents an attention block, which includes multihead ProbSparse self-attention and necessary operations. And $\text{Conv1d}(\cdot)$ uses the $\text{ELU}(\cdot)$ activation function to perform a 1-D convolution filtering (kernel width = 3) in the time dimension. A max-pooling layer is added after the convolution operation, followed by the downsampling of X^t , which reduces overall memory usage to $O((2 - \epsilon)L \log L)$, with ϵ being a small value [2].

D. Implementation of EEGNet-MSD Decoder

In this study, we employed two essential modules of the EEGNet model proposed by Lawhern et al. [11] to implement our model. For features extraction, a flattened layer was included to reshape the output, fed into a multiheaded ProbSparse self-attention module to produce a feature matrix with updated weights. Then, the feature matrix was connected to the self-attention extraction layer, and the resulting output was then passed to a classification layer that utilized the Softmax activation function for classification. The specific structure of the EEGNet-MSD decoder is illustrated in Fig. 5. Set the values in Fig. 5 based on BCI Competition IV dataset 2a.

TABLE I

HYPERPARAMETER VALUES FOR THE EEGNET DECODER

Hyperparameter	Values
LK	16, 32, 64, 128
F1	4, 6, 8, 16
F2	4, 6, 8, 16
D	4, 6, 8
Optimizer	Adam, RMSprop, SGD
Learning rate	0.0001, 0.001, 0.01
Batch size	16 32 64 128
Regularization of L2	0.01, 0.001
Dropout	0.2, 0.3, 0.5
Epochs	300 400 800

E. Data Preprocessing**1) Data Preprocessing for BCI Competition IV Dataset 2a:**

The performance of decoders can be enhanced through preprocessing of EEG signals. The EEG signals of 22 channels were selected from the dataset and the EOG signals were removed, after which they were processed using 7–35-Hz bandpass filtering and no filtering, respectively. Then, for each trial, a data sample of 2–6 s was selected because the data reflecting the subject's MI manipulation appeared within this time frame in the 7.5-s sample duration. The data were standardized and scaled to a range of 0–1. Subsequently, the one-hot encoding was used for the four classes of MI to form 4-D vectors as: $\{[1, 0, 0, 0], [0, 1, 0, 0], [0, 0, 1, 0], [0, 0, 0, 1]\}$.

2) Data Preprocessing for Physiobank EEG Motor Movement/Imagery Dataset: The EEG signals from the Physiobank dataset were filtered using a bandpass filter with a frequency range of 7–35 Hz. To facilitate comparison with other studies, MI data, imagining the left and right wrist movements were selected in the Physiobank dataset for binary classification experiments. The EEG signals of 64 channels were selected from the dataset. Then, for each trial, a 4-s data sample was chosen because each MI lasted for 4 s during the experiment. The train set included data from subject 1 (S001) to subject 84 (S084), and the remaining subjects were used in the test set. Based on the studies of Varsehi and Firoozabadi [27] and Fan et al. [28], due to samples missing in four subjects, 88 (S088), 92 (S092), 100 (S100), and 104 (S104), the relevant data were removed from the test set. To perform binary classification, the one-hot encoding method was used to represent the two types of MI involving the left and right wrist as 2-D vectors: $\{[1, 0], [0, 1]\}$. The brain signals were normalized in a range of 0–1. During the training phase, 80% of the data in the dataset is used for training, while the remaining data is used for testing.

F. Hyperparameters Optimization of Decoders

For the EEGNet neural network model, the hyperparameter was optimized using data from the third subject in the BCI Competition IV dataset 2a. After obtaining the best results in this participant's test set, the model's hyperparameter settings were saved and applied to train the optimal model on other participants' datasets. The decoding hyperparameter settings for EEGNet are shown in Table I.

TABLE II

HYPERPARAMETER VALUES OF THE EEGNet-MSD DECODER FOR THE BCI COMPETITION IV DATASET 2A

Hyperparameter	Values
n-heads	2, 4, 6, 8, 16
reshape size	128, 256, 384, 512, 1024

TABLE III

HYPERPARAMETER VALUES OF THE EEGNet-MSD DECODER FOR THE PHYSIOBANK EEG MOTOR MOVEMENT/MI DATASET

Hyperparameter	Values
n-heads	8, 16
Batch size	64, 128, 256

To obtain the optimal hyperparameter settings, this study performed hyperparameter optimization on the EEGNet-MSD decoder using the EEG data of the third subject in the BCI Competition IV dataset 2a. The hyperparameter values of the best model were saved and applied to the EEG data of other subjects. The hyperparameters specific to the EEGNet-MSD decoder are shown in Table II, while the hyperparameter settings shared with EEGNet are shown in Table I.

Due to the large number of subjects in the Physiobank EEG motor movement/MI dataset, adjustments were made for the batch size hyperparameter. The specific hyperparameter values are given in Table III.

III. RESULTS

A. Optimal Hyperparameters for the EEGNet-MSD Decoders

After arranging and combining all the possible hyperparameter combinations, we conducted the training and testing phases. Table IV shows the best hyperparameter settings for both datasets. The EEGNet-MSD decoder was optimized on the BCI Competition IV dataset 2a, and the optimal hyperparameters were obtained. The specific parameter values of EEGNet were set to $F1 = 8$, $D = 8$, $F2 = 16$, $LK = 64$, and eight heads. For training on the PhysioBank EEG motor/imagery dataset, the batch size was adjusted to 64 and the number of epochs to 300 to achieve the best test results.

B. Comparison Between the Implemented Decoders Using BCI Competition IV Dataset 2a

The classification performance was assessed by testing the validation data provided in the competition after training each participant's data with the optimal hyperparameter settings. The accuracy comparison between the EEGNet-MSD decoder and LSTM neural decoder on the test set is shown in Fig. 6.

The results showed that the EEGNet-MSD decoder achieved higher or comparable accuracy in testing data from the remaining subjects compared to the LSTM neural decoder. In addition, this study compared the kappa values obtained by the EEGNet-MSD and LSTM neural decoders, as shown in Fig. 7. A bar graph was used to show the kappa values. Overall, the EEGNet decoder performed better than the LSTM neural decoder.

TABLE IV

OPTIMAL HYPERPARAMETER SETTINGS FOR THE EEGNet-MSD DECODER FOR EACH CORRESPONDING DATASET

Hyperparameter	BCI Competition IV dataset 2a	Physiobank EEG motor movement/imagery dataset
LK	64	64
F1	8	8
F2	16	16
D	8	8
Optimizer	Adam	Adam
Learning rate	0.001	0.001
Batch size	64	16
Regularization of L2	0.001	0.001
Dropout	0.3	0.3
Epochs	400	300
n-heads	8	8

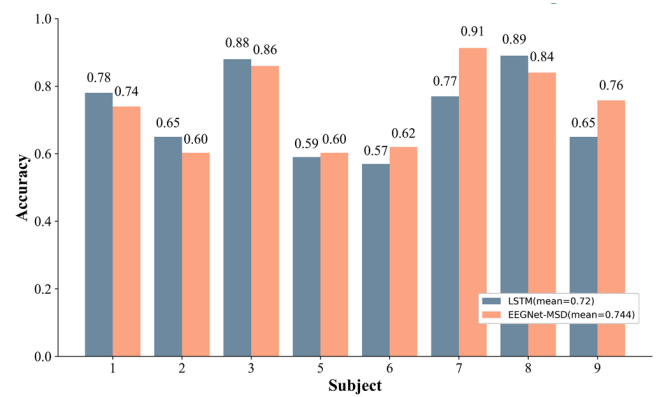


Fig. 6. Comparison of the accuracy obtained through the decoders.

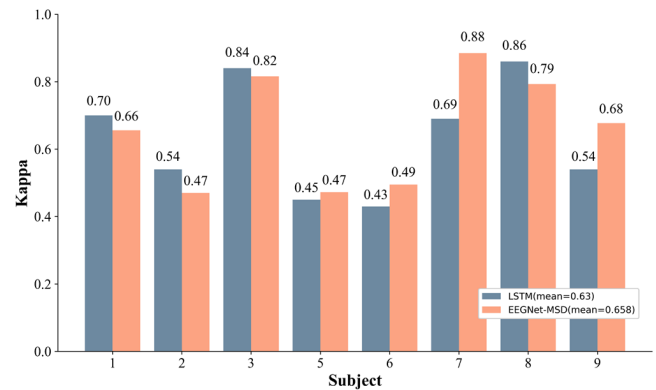


Fig. 7. Comparison of the kappa obtained through the decoders.

Compare the kappa values obtained by the EEGNet-MSD decoder with those obtained by other decoders in previous studies and the results are shown in Fig. 8. The EEGNet-MSD decoder shows strong consistency [29] with an average kappa value of 0.66. The filter-bank spatial filtering and temporal-spatial CNN (FBSF-TSCNN) [30] decoder and spatial compact CNN and recurrent neural network (SCCRNN) [31] made use of CNNs, with an average kappa value of 0.64. The filter bank common spatial pattern (FBCSP) and namely batch and pairwise (NBPW) [32] decoder ranked first in the BCI Competition IV, achieving an average kappa value

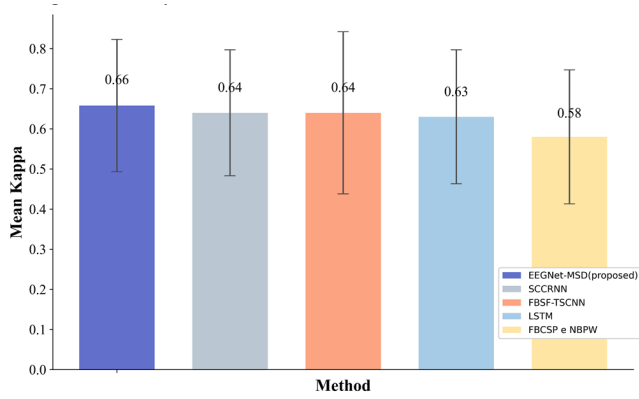


Fig. 8. Comparison of mean kappa and std obtained through the decoders.

TABLE V

COMPARISON OF TIME COMPLEXITY AND SPACE COMPLEXITY

Methods	Training		Testing
	Time	Memory	Steps
EEGNet-MSD	$O(L \log L)$	$O(L \log L)$	1
LSTM	$O(L)$	$O(L)$	L

TABLE VI

COMPARISON OF AVERAGE ACCURACY VALUES OBTAINED IN OTHER STUDIES

Methods	FBCSP&SVM	EEGNet	LSTM	EEGNet-MSD
Accuracy value	0.64	0.82	0.85	0.86

of 0.58. The LSTM neural decoder, with an average kappa value of 0.63, uses wavelet packet decomposition (WPD) and common spatial patterns (CSPs) for feature extraction and uses LSTM layers for classification. The comparison shows that the proposed method has good stability on the basis of high accuracy.

Based on accuracy comparison, the time and space complexity during training of EEGNet-MSD decoder with self-attention mechanism layer and LSTM neural decoder were calculated and shown in Table V, which shows that the EEGNet-MSD decoder has significantly lower time and space complexity than the LSTM neural decoder.

C. EEGNet-MSD Decoder Performance With Physiobank EEG Motor Movement/Imagery Dataset

The EEGNet-MSD decoder was trained and tested on the Physiobank EEG motor movement/MI dataset, achieving an average classification accuracy of 0.86. As shown in Table VI, our result is higher than the accuracy of 0.82 attained by EEGNet [33] and 0.85 attained by LSTM neural decoder, and Handiru and Prasad [34] only managed a 0.64 accuracy using an FBCSP and support vector machine.

IV. DISCUSSION

The study presents the EEGNet-MSD decoder, which maintains high accuracy and reduces the time and space complexity of decoding through a sparse self-attention mechanism. The ProSparse self-attention mechanism is used as a reference, and

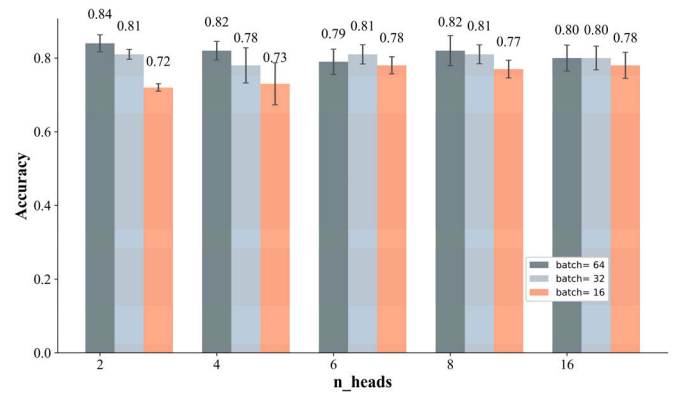


Fig. 9. Decoding accuracy corresponding to the number of heads in different batch sizes.

a sampling factor is implemented to reduce dot product calculations. A sparsity measure is used to update importance with weights, while nonimportant features are directly averaged to decrease computation and maximize the utilization of multiple channels. This sparse selection mechanism is used after the two feature extraction modules of EEGNet, and the sampling factor c is a constant between 0 and 1, which determines the number of sparse selected features. Additionally, a max pooling layer is introduced to reduce the length of the time dimension and computational complexity. The EEGNet-MSD decoder was evaluated on BCI Competition IV dataset 2a, and an accuracy of approximately 8% higher than the best result in Competition IV was achieved. Compared to the latest LSTM neural decoder, the EEGNet-MSD decoder is superior in terms of time and space complexity and classification accuracy. On the Physiobank EEG motor/imagery dataset, the EEGNet-MSD decoder has an accuracy of 0.86, which is higher than the LSTM neural decoder with an accuracy of 0.85 and EEGNet [31] with an accuracy of 0.82. This indicates that EEGNet-MSD effectively solves complex problems and can be replicated. However, when using EEGNet as a decoder, its classification accuracy is only 0.82. This suggests that using EEGNet alone as a decoder may not fully leverage the relevant information within each channel. Therefore, it is worth considering the use of a multihead self-attention mechanism to improve classification accuracy. We explored the impact of the number of attention heads on decoding accuracy under different batch sizes. The MI data of subject 3 in the competition IV dataset was used for training, with batch sizes of 16, 32, and 64, respectively. As shown in Fig. 9, with a batch size of 64, the result in 2 and 8 heads is more accurate and stable, indicating the importance of selecting an appropriate number of heads for decoding accuracy. In this study, we combined the accuracy of decoders with different batch sizes and ultimately selected eight heads in the attention module.

Due to the short duration of valid EEG signals in the dataset used in this study, the number of features input into the neural network is limited. Hence, only one attention mechanism module is utilized to alleviate the loss of feature information in the self-attention mechanism. To decode extended time sequence signals, it might be worthwhile to contemplate

enhancing the attention modules within the decoder, similar to those used in online EEG decoding. In the BCI Competition data preprocessing stage, the result of using bandpass filtering 7–35 Hz contrasted with those of no filtering, and it was indicated by the results that the performance of the EEGNet-MSD decoder was even better without using any filtering. Satisfactory results were obtained by applying bandpass filtering between 7 and 35 Hz and selecting μ and β frequency ranges to decode and classify two classes of MI from the PhysioBank EEG MI dataset. The traditional signal decomposition and feature extraction methods for EEG are generally more complex [35], [36]. The process to extract features from EEG signals in time-frequency and spatial domains using WPD [37] and CSP [38] is separate from the LSTM neural network in Tian's (2023) LSTM neural decoder. The signals were analyzed by WPD, which allowed for examining their time and frequency attributes. The CSP technique represented the signals at various frequency-time scales and extracted spatial properties. Due to this separation, uncertainty arises during the decoding process and may result in the inability to promptly adjust feature extraction based on classification results, leading to suboptimal solutions. Therefore, the EEGNet-MSD decoder integrates feature extraction, attention processing, and classification stages and modifies feature extraction and attention processing stages based on classification results to improve decoding accuracy.

V. CONCLUSION

In this study, we proposed the EEGNet-MSD decoding approach to efficiently extract features from EEG signals and accurately classify EEG-based intent. In terms of decoder structure, two feature extraction modules of EEGNet were combined with a sparse multihead attention mechanism, which improves the decoding accuracy. The sparse attention mechanism was used to reduce time and spatial complexity, and the attention distillation mechanism was introduced to further reduce computational costs. To obtain the optimal hyperparameters for the decoder, we designed some combinations for hyperparameter selection and applied them to the training of the decoder. The results showed that the self-attention mechanism, hyperparameter selection, and data preprocessing all affected the performance of the EEGNet-MSD decoder. The sparse selection method of feature weights used reveals more feature information in convolutional networks, increases the interpretability of the model, and provides a solution for explaining neural networks in attention interpretation. Moreover, our model reduces the computational complexity of decoding, and future research should focus more on using this method in the field of online EEG intent decoding.

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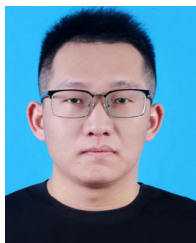
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